

COMP 2310 - Theoretical Foundations of CS

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office h.: Mo We (11:30-12:30), We (13-15)

labs start from 3rd week (Mon Jan 19)

Contents: logic, binary relations, graphs, recurrences

Book (optional): Discrete Math (K. H. Rosen)

Evaluation: - 3 labs (30%)

- midterm (tentative: We, Feb 25, 10-11:20) 30%

- Final 40%

registration/access to Brightspace (csinfo@uwindsor.ca)

Logic (review)

proposition: a statement that is T or F

- Windsor is capital of Canada ✓
- Today is Monday ✓
- what time is it? x

logical operators

negation $\sim, \neg$

$\sim p$ ^{prop.}

conjunction (and) $p \wedge q$

disjunction (or) $p \vee q$

biconditional $p \Leftrightarrow q$

this is true if p and q have the same value

implication
(conditional)

$p \Rightarrow q$

it says if p is true then q is true

$P \Rightarrow Q$:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

I did not lie

I lied

I did not lie

I did not lie

P: you get ≥ 80 in the course

Q: I give you A+

n variables / atomic propositions $\Rightarrow 2^n$ rows in truth table

p, q ($n=2$) $\Rightarrow 2^2 = 4$ rows

Priority of operators:

$\sim, \neg$
 $\wedge$
 $\vee$
 $\Rightarrow$
 $\Leftrightarrow$

↑
higher
priority

$P \Rightarrow q \wedge \sim p \vee r \wedge s$

$(P \Rightarrow (q \wedge (\sim p)) \vee (r \wedge s))$
 5 2 1 4 3

tautology: a statement that is always True

contradiction: " " " " " False

p	q	r	X	Y
—	—	—	T	F
—	—	—	T	F
—	—	—	T	F
—	—	—	T	F

X is tautology

Y is contradiction

Equivalency: We say A and B are equivalent, $A \equiv B$,

if they have same values in all the rows

(for all truth assignments)

exmp: $A \equiv B$
 $p \Rightarrow q \equiv \sim p \vee q$

p	q	$p \Rightarrow q$	$\sim p \vee q$	$(p \Rightarrow q) \Leftrightarrow (\sim p \vee q)$
T	T	T	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$A \equiv B$ if and only if $A \Leftrightarrow B$ is a tautology

this statement is a tautology

Some known logical equivalences: α, β, γ

$$\begin{aligned} \alpha \wedge T &\equiv \alpha \\ \alpha \vee F &\equiv \alpha \end{aligned} \quad \left. \vphantom{\begin{aligned} \alpha \wedge T &\equiv \alpha \\ \alpha \vee F &\equiv \alpha \end{aligned}} \right\} \text{identity}$$

$$\alpha \wedge F \equiv F \quad \left. \vphantom{\alpha \wedge F \equiv F} \right\} \text{domination}$$

$$\alpha \wedge \alpha \equiv \alpha \quad \left. \vphantom{\alpha \wedge \alpha \equiv \alpha} \right\} \text{idempotent}$$

$$\sim(\sim\alpha) \equiv \alpha \quad \left. \vphantom{\sim(\sim\alpha) \equiv \alpha} \right\} \text{double negation}$$

$$\alpha \vee \beta \equiv \beta \vee \alpha \quad \left. \vphantom{\alpha \vee \beta \equiv \beta \vee \alpha} \right\} \text{commutative}$$

$$\alpha \vee (\beta \vee \gamma) \equiv (\alpha \vee \beta) \vee \gamma \quad \left. \vphantom{\alpha \vee (\beta \vee \gamma) \equiv (\alpha \vee \beta) \vee \gamma} \right\} \text{associative}$$

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma) \quad \left. \vphantom{\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)} \right\} \text{distributive}$$

$$\sim(\alpha \vee \beta) \equiv \sim\alpha \wedge \sim\beta \quad \left. \vphantom{\sim(\alpha \vee \beta) \equiv \sim\alpha \wedge \sim\beta} \right\} \text{De Morgan}$$

$$\alpha \wedge (\alpha \vee \beta) \equiv \alpha \quad \left. \vphantom{\alpha \wedge (\alpha \vee \beta) \equiv \alpha} \right\} \text{absorption}$$

$$\alpha \wedge \sim\alpha \equiv F \quad \left. \vphantom{\alpha \wedge \sim\alpha \equiv F} \right\} \text{negation}$$

$$\alpha \vee \sim\alpha \equiv T$$

$$\alpha \Rightarrow \beta \equiv \sim\alpha \vee \beta \quad \text{Implication}$$

$$\alpha \Rightarrow \beta \equiv \sim\beta \Rightarrow \sim\alpha \quad \text{Contraposition / contrapositive}$$

exmp: Prove that if $\underbrace{n \text{ is even}}_{\alpha}$ then $\underbrace{n^2 \text{ is even}}_{\beta}$

$$\alpha \Rightarrow \beta$$

$$\equiv$$

$$\sim\beta \Rightarrow \sim\alpha$$

contraposition: Prove that $\sim\beta \Rightarrow \sim\alpha$

prove that if n^2 is odd then n is odd

Theorem: a statement with two parts

Theorem T:

① Hypotheses $H_1, H_2, \dots, H_k$

② Conclusion C

A theorem is correct if hypotheses imply the conclusion
if $H_1, H_2, \dots$ and H_k imply C

$$H_1 \wedge H_2 \wedge \dots \wedge H_k \Rightarrow C$$



To prove theorem T we need to show this is always true
(is a tautology)

Formal proof: A sequence of valid statements that lead to C

valid statements are:

① Hypotheses $H_1, H_2, \dots, H_k$

② tautology

③ obtained by logical equivalences

④ obtained by inference rules

will see this later

Theorem: If $\underbrace{p \Rightarrow q}_{\text{Hyp.}}$ then $\underbrace{p \Rightarrow (p \wedge q)}_{\text{Conc.}}$.

Prove: 1. $p \Rightarrow q$ Hyp.

2. $\sim p \vee q$ Implication ①

3. $(\sim p \vee q) \wedge T$ Identity ②

$$\alpha \wedge T \equiv \alpha$$

4. $T \wedge (\sim p \vee q)$ Commutative ③

5. $(\sim p \vee p) \wedge (\sim p \vee q)$ negation ④

6. $\sim p \vee (p \wedge q)$ distributive ⑤

7. $p \Rightarrow (p \wedge q)$ Implication ⑥

$$\alpha \Rightarrow \beta \equiv \sim \alpha \vee \beta$$

✓